



file. In our comparison of codecs, we talk about four widely-used codecs that are supported by most media players: DivX 5.1.1, QuickTime 6.5 / Sorenson3, Windows Media Video 9, and MPEG-4. DivX videos are widely available online, as are QuickTime videos. WMV 9 encoded video is playable using Windows Media Player, and is playable on many PMPs and DVD players. MPEG-4 is compatible with consumer electronic devices like the PSP, Xbox 360, and the Apple iPod.

A typical video clip has a variety of images—fast paced action scenes, close-ups, slow motion etc. We compared four codecs in different test cases. In our first case, the DVD-quality source video, which is encoded using MPEG2, was a scene from *The Matrix: Reloaded*, with lots of fast action scenes.

In this video, the output from DivX is almost as good as the original, while the output from MPEG-4 video is the worst, with blocky images. You would have to look closely, though, for the differences to be noticeable. In any case, these differences are not very visible in motion because persistence of vision, but a good codec should not display distortion in still shots either.

We also looked at some other clips: the Green Goblin's attack on the balloon parade in *Spider-Man*, Nightcrawler's attack on the President in *X-Men 2*, and the two monsters Mike and Sully coming out for work in the morning in *Monsters, Inc.* Anybody who remembers these scenes would know that they do not always depict slow moves and gentle camera motion; there is also a lot of fast-paced action. Space doesn't permit all the screenshots, but from the decoded output in the different cases, the DivX and WMV 9 videos were almost tied, while the MPEG-4 encoding fared the worst. The DivX-encoded videos had a more detailed and sharper look, but also displayed blocks in the output. WMV 9 blanks out details, and the videos have a softer look. When it comes to speed, MPEG-4 and QuickTime / Sorenson was pretty fast, with DivX at third place.

Encoding video takes a lot of time in any case—a 4:16 minute clip of *The Matrix Reloaded* at a bitrate of 1 Mbps took 13:15 minutes with DivX, 17:22 minutes with WMV9, 8:37 minutes with QuickTime / Sorenson, and 11:50 minutes with MPEG-4.

The choice of codec comes down to the intended use of your video: if you plan on putting it up on the Net, go with DivX; WMV goes very well with devices like DVD players, and has wider industry support.

And So...

Video compression is a vast topic, and the technical details can fill several fat books. We've only attempted here at giving a peek into what goes on behind the scenes.

Compression saves space, but the choice of codec is more important than the space the output file will occupy. Also, the codec you use should be widely supported. ■

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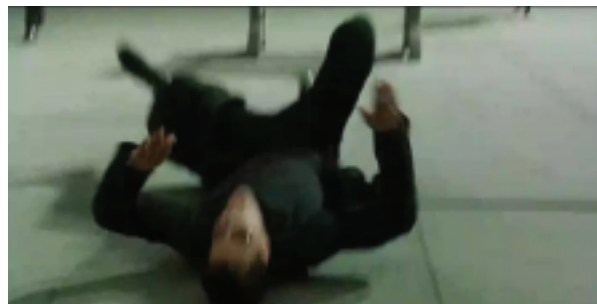
This is the original video grab. The image is blurry in some areas.



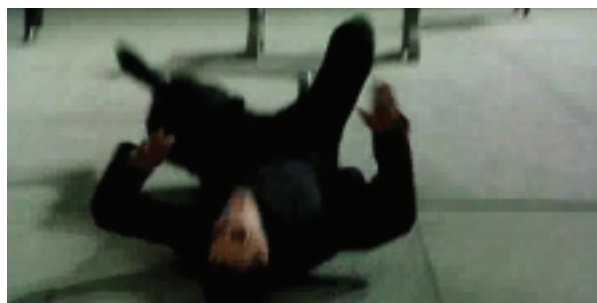
The DivX video gives the best output, almost as good as the original



The WMV video blurs out the details, and edges are jagged



The blockiness starts in QuickTime—notice the shadows and edges



The MPEG-4 gives the worst output. The image is blocky, and there is not much detail